STA2002: Probability and Statistics II
Tests of Statistical Hypotheses–Part 2

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Introduction

- In this lecture we will continue our journey in hypothesis testing and discuss hypothesis testing for equality of two means and for equality of two variances.
- In particular, we will learn the following tests today:
  - pooled t-test
  - Welch's t-test
  - paired t-test
  - F-test
- **Suggested reading**: Chapter 8.1, 8.2 of the textbook.

Recap of hypothesis Testing Framework

1. Formulate H_0 and H_1, and determine the significance level α, say $\alpha = 0.05$ or 0.01.
2. Determine the test statistic T, its distribution under H_0 and compute its realized value t based on the data.
3. Draw conclusion (i.e. reject or fail to reject H_0) based on the following equivalent conditions:

   (a) Reject H_0 if p-value $\leq \alpha$. Otherwise, we fail to reject H_0.

   (b) Reject H_0 if $t \in C$, where C is the critical region. Otherwise, we fail to reject H_0.

   (c) confidence interval approach.

Hypothesis Testing for Equality of Two Normal Means

- Suppose that we have two populations from which we draw i.i.d. random samples, say $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$.

- We would like to test

$$H_0 : \mu_X = \mu_Y$$

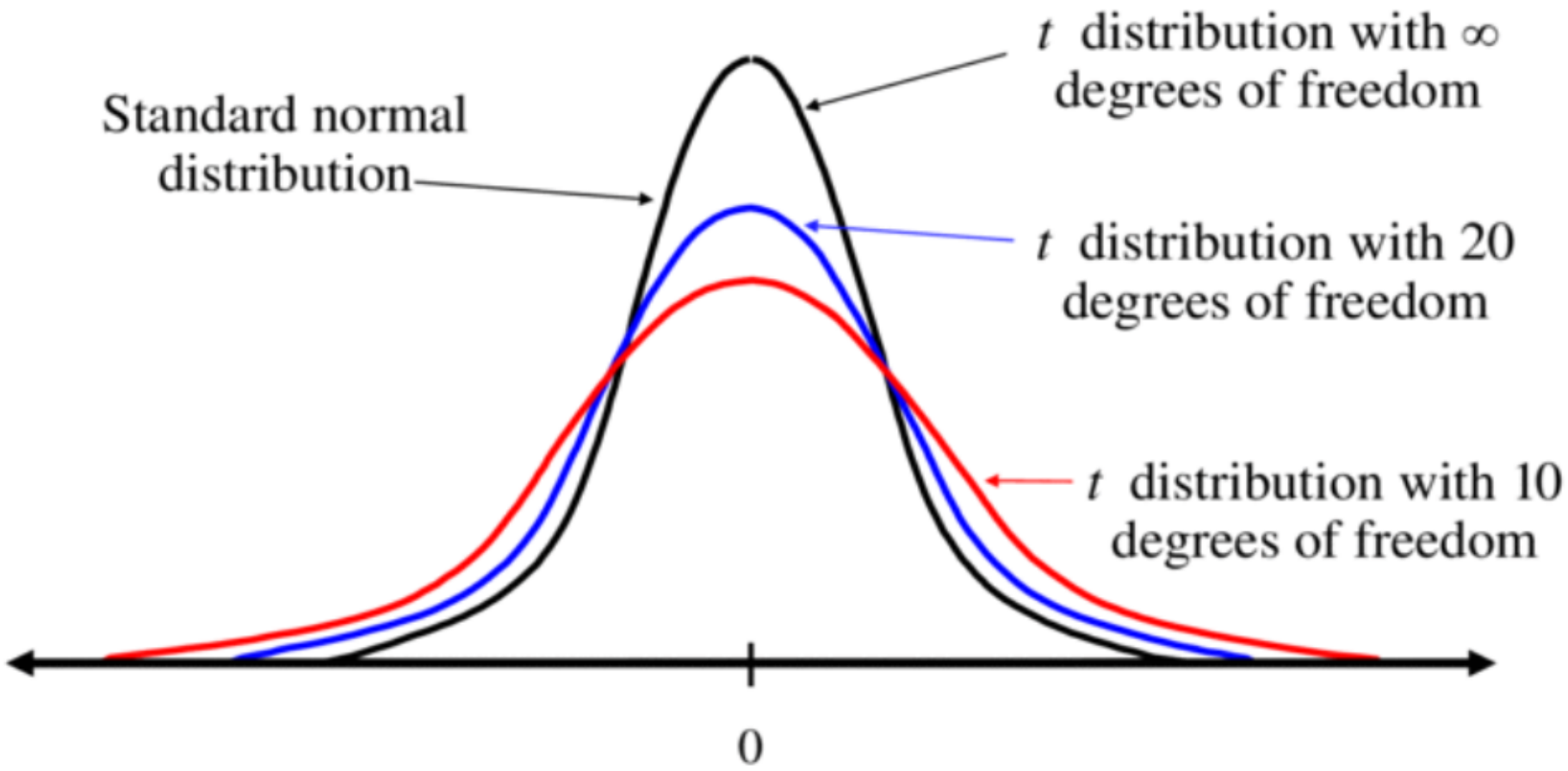
against some possible competing H_1.

Hypothesis Testing for Equality of Two Normal Means

We will discuss 3 cases:

1. Case 1: $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are independent
   1. **(pooled t-test)** $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, and σ^2 is unknown (Two populations have the same variance σ^2)
   2. **(Welch's t-test)** $\sigma_X^2 \neq \sigma_Y^2$, and they are unknown
2. **(paired t-test)** Case 2: $n = m$ and X_i, Y_i are dependent for $i = 1, \ldots, n$. However, the n pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent.

t-Distribution



Case 1(a): Pooled t-test:
(One-sided test) $H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X > \mu_Y$

- We consider the pooled statistic

$$T := \frac{\overline{X} - \overline{Y}}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}},$$

where we recall that S_p^2 is the pooled estimator of σ^2,

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2 + \sum_{i=1}^m (Y_i - \overline{Y})^2}{n+m-2}.$$

recall: sample variance $S^2 = \frac{1}{n-1}\sum (X_i - \overline{X})^2$

- Under H_0, $T \sim t(n+m-2)$.
- Once we observe the data $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$, we can compute

$$\boxed{t = \frac{\overline{x} - \overline{y}}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}.}$$

Case 1(a): Pooled t-test:
(One-sided test) $H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X > \mu_Y$

- p-value is

$$p = P(T > t; H_0) = P(t(n+m-2) > t),$$

so we reject H_0 at significance level α when

$$\underbrace{\boxed{\text{p-value} = P(t(n+m-2) > t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \geq t_\alpha(n+m-2)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{0 \notin [\overline{x} - \overline{y} - t_\alpha(n+m-2)s_p\sqrt{\frac{1}{n}+\frac{1}{m}}, \infty)}}_{\text{confidence interval approach}}$$

Case 1(a): Pooled t-test:
(One-sided test) $H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X < \mu_Y$

- p-value is

$$p = P(T < t; H_0) = P(t(n+m-2) < t),$$

so we reject H_0 at significance level α when

$$\underbrace{\boxed{\text{p-value} = P(t(n+m-2) < t) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{t \leq -t_\alpha(n+m-2)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{0 \notin (-\infty, \overline{x} - \overline{y} + t_\alpha(n+m-2)s_p\sqrt{\frac{1}{n}+\frac{1}{m}}]}}_{\text{confidence interval approach}}$$

Case 1(a): Pooled t-test:
(Two-sided test) $H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X \neq \mu_Y$

- p-value is

$$p = P(|T| > |t|; H_0) = P(t(n+m-2) > |t|) + P(t(n+m-2) < -|t|)$$
$$= 2P(t(n+m-2) > |t|),$$

so we reject H_0 at significance level α when

$$\underbrace{\boxed{\text{p-value} = 2P(t(n+m-2) > |t|) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{|t| \geq t_{\alpha/2}(n+m-2)}}_{\text{critical region approach}} \qquad t_{\alpha/2} := t_{\alpha/2}(n+m-2)$$

$$\Leftrightarrow \quad \underbrace{\boxed{0 \notin [\overline{x} - \overline{y} - t_{\alpha/2} s_p \sqrt{\frac{1}{n} + \frac{1}{m}}, \overline{x} - \overline{y} + t_{\alpha/2} s_p \sqrt{\frac{1}{n} + \frac{1}{m}}]}}_{\text{confidence interval approach (pooled t-interval)}}$$

Hypothesis Testing for Equality of Two Normal Means

Table 8.2-1 Tests of hypotheses for equality of two means

H_0	H_1	Critical Region
$\mu_X = \mu_Y$	$\mu_X > \mu_Y$	$t \geq t_\alpha(n+m-2)$ or $\bar{x} - \bar{y} \geq t_\alpha(n+m-2)s_p\sqrt{1/n + 1/m}$
$\mu_X = \mu_Y$	$\mu_X < \mu_Y$	$t \leq -t_\alpha(n+m-2)$ or $\bar{x} - \bar{y} \leq -t_\alpha(n+m-2)s_p\sqrt{1/n + 1/m}$
$\mu_X = \mu_Y$	$\mu_X \neq \mu_Y$	$\lvert t\rvert \geq t_{\alpha/2}(n+m-2)$ or $\lvert\bar{x} - \bar{y}\rvert \geq t_{\alpha/2}(n+m-2)s_p\sqrt{1/n + 1/m}$

Example: Example 8.2-2

Example 1

$X \sim N(\mu_X, \sigma^2)$ and $Y \sim N(\mu_Y, \sigma^2)$. We would like to test

$$H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X \neq \mu_Y,$$

at significance level $\alpha = 0.10$. We have
$n = 12, \overline{x} = 1076.75, s_X^2 = 29.30$ and
$m = 12, \overline{y} = 1072.33, s_Y^2 = 26.24$. We can then calculate

$$t = \frac{1076.75 - 1072.33}{\sqrt{\frac{11(29.30)+11(26.24)}{22}\left(\frac{1}{12}+\frac{1}{12}\right)}} = 2.05$$

$$t_{0.05}(22) = 1.717$$

two side. $\alpha = 10\%$

Example: Example 8.2-2

Solution.
critical region approach:

$$|t| = 2.05 > 1.717,$$

and so we reject H_0 at 10% significance level.
p-value approach:

$$p = 2P(t(22) > 2.05) = 0.0525 < 0.1,$$

and so we reject H_0 at 10% significance level. However, we fail to reject H_0 at 5% significance level. □

Case 1(b): Welch's t-test

- We shall only consider two-sided hypothesis testing. The derivation for one-sided test is similar and is omitted here.
- Consider the following two-sided test,

$$H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X \neq \mu_Y$$

- We consider the Welch's statistic,

$$T := \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}.$$

Case 1(b): Welch's t-test

- Under H_0, $T \sim t(r)$ *approximately*, and we recall that,

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor.$$

- Once we observe the data $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$, we can compute,

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}.$$

Case 1(b): Welch's t-test

- p-value is

$$p = P(|T| > |t|; H_0) = 2P(t(r) > |t|),$$

so we reject H_0 at significance level α when

$$\underbrace{\boxed{\text{p-value} = 2P(t(r) > |t|) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{|t| \geq t_{\alpha/2}(r)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{0 \notin [\overline{x} - \overline{y} - t_{\alpha/2}(r)\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}, \overline{x} - \overline{y} + t_{\alpha/2}(r)\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}]}}_{\text{confidence interval approach (Welch's t-interval)}}$$

Case 2: paired t-test

- We shall only consider two-sided hypothesis testing. The derivation for one-sided test is similar and is omitted.
- Consider the following two-sided test,

$$H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X \neq \mu_Y$$

- Introduce D_i, $i = 1, 2, \ldots, n$,

$$D_i := X_i - Y_i$$

- We consider the paired statistic,

$$T := \frac{\overline{D}}{S_D/\sqrt{n}} = \frac{\overline{X} - \overline{Y}}{S_D/\sqrt{n}}.$$

Case 2: paired t-test

- Under $H_0: \mu_X = \mu_Y$, $T \sim t(n-1)$, see previous lecture slides and recall the Student's Theorem; or read Section 7.2 of the textbook.

- Once we observe the data $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$, we can compute,

$$\boxed{t = \frac{\overline{x} - \overline{y}}{s_D / \sqrt{n}}.}$$

- p-value is

$$p = P(|T| > |t|; H_0) = 2P(t(n-1) > |t|).$$

Case 2: paired t-test

- So we reject H_0 at significance level α when

$$\underbrace{\boxed{\text{p-value} = 2P(t(n-1) > |t|) \leq \alpha}}_{\text{p-value approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{|t| \geq t_{\alpha/2}(n-1)}}_{\text{critical region approach}}$$

$$\Leftrightarrow \quad \underbrace{\boxed{0 \notin [\overline{x} - \overline{y} - t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}, \overline{x} - \overline{y} + t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}]}}_{\text{confidence interval approach (paired t-interval)}}$$

Example: Example 8.1-5

Example 2

We would like to test

$$H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X \neq \mu_Y,$$

at significance level $\alpha = 0.10$. We have
$n = 24, \overline{x} - \overline{y} = 0.0788, s_D = 0.2549$. We can then calculate

$$t = \frac{0.0788}{0.2549/\sqrt{24}} = 1.514$$

$$t_{0.05}(23) = 1.714$$

Example: Example 8.1-5

Solution.
critical region approach:

$$|t| = 1.514 < 1.714,$$

and so we fail to reject H_0 at 10% significance level.
p-value approach:

$$p = 2P(t(23) > 1.514) = 0.1436 > 0.1,$$

and so we fail to reject H_0 at 10% significance level.
confidence interval approach: The 90% two-sided paired C.I. is

$$0.0788 \pm 1.714 \times \frac{0.2549}{\sqrt{24}} = [-0.01, 0.17],$$

which includes 0, and so we fail to reject H_0 at 10% significance level. □

Hypothesis Testing for Equality of Two Normal Variances

- Suppose that we have two populations from which we draw i.i.d. random samples, say $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$.
- Also, assume that they are independent. We would like to test

$$H_0 : \sigma_X^2 = \sigma_Y^2 \text{ v.s. } H_1 : \sigma_X^2 \neq \sigma_Y^2.$$

- Recall the definition of *F*-distribution: If $U \sim \chi^2(u)$ and $V \sim \chi^2(v)$ and they are independent, then the ratio

$$\frac{U/u}{V/v} \sim F(u, v).$$

Hypothesis Testing for Equality of Two Normal Variances

- Now, we take

$$U = \frac{(n-1)S_X^2}{\sigma_X^2} \sim \chi^2(n-1),$$

$$V = \frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi^2(m-1),$$

  and note that they are independent (since X and Y are independent).
- The following ratio is constructed,

$$\frac{U/u}{V/v} = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1).$$

Hypothesis Testing for Equality of Two Normal Variances

- Let the test statistic T be,

$$T := \frac{S_X^2}{S_Y^2}.$$

- Under $H_0: \ \sigma_X^2 = \sigma_Y^2$,

$$T = \frac{S_X^2}{S_Y^2} = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1).$$

- Once we have the data $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$, we calculate

$$\boxed{t = \frac{s_X^2}{s_Y^2}}.$$

Hypothesis Testing for Equality of Two Normal Variances

- Intuitively, the data reject H_0 and favours H_1 when the ratio t is either too small or too large.
- Precisely, we reject H_0 at significance level α when

$$\underbrace{t \geq F_{\alpha/2}(n-1, m-1) \quad \text{or} \quad t \leq F_{1-\alpha/2}(n-1, m-1)}_{\text{critical region approach}}$$

- This is called ***F*-test**.
- **Remark**. For F-distribution,

$$F_{1-\alpha/2}(n-1, m-1) = \frac{1}{F_{\alpha/2}(m-1, n-1)},$$

which is useful when we use the F-distribution table.

F-Distribution

